THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE UNIT TEST SET 3

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The molecular mass of H_2O is (H = 1, O = 16): 1

(a) 16 u **(b)** 18 u **(c)** 20 u **(d)** 17 u

2. An atom has 11 protons and 12 neutrons. Its mass number is: 1

(a) 34 **(b)** 23 **(c)** 11 **(d)** 12

3. An example of a chemical change is: 1

(a) tearing paper **(b)** melting of ice
(c) burning of magnesium **(d)** dissolving sugar

4. Alloys are: 1

(a) pure metals **(b)** heterogeneous
(c) homogeneous mixtures of metals **(d)** compounds

5. The formula of calcium chloride is: 1

(a) $CaCl$ **(b)** Ca_2Cl **(c)** Ca_2Cl_2 **(d)** $CaCl_2$

6. Isotopes are atoms of the same element with the same: 1

(a) number of nucleons **(b)** mass number but different atomic numbers
(c) atomic number but different mass numbers **(d)** number of neutrons

7. A well-tested scientific theory: 1

(a) is a measurement
(b) explains many observations and makes testable predictions
(c) can never be changed
(d) is just a guess

8. A mixture of sand and iron filings is: 1

(a) heterogeneous **(b)** homogeneous **(c)** a compound **(d)** a pure substance

9. 12 g of carbon burns completely in 32 g of oxygen. The mass of carbon dioxide formed is: 1

(a) 28 g **(b)** 20 g **(c)** 12 g **(d)** 44 g

10. The neutron was discovered by: 1

(a) James Chadwick **(b)** Bohr **(c)** Rutherford **(d)** Thomson

11. An aircraft ran out of fuel because of a mix-up of units, showing that: 1

(a) units do not matter
(b) units must be used carefully and consistently
(c) fuel is not measurable
(d) planes need no fuel

12. The separation of cream from milk is done by: 1

(a) centrifugation **(b)** distillation **(c)** sublimation **(d)** filtration

13. The molecular mass of CO_2 is (C = 12, O = 16): 1

(a) 56 u **(b)** 44 u **(c)** 28 u **(d)** 32 u

14. The number of neutrons in $^{35}_{17}Cl$ is: 1

(a) 18 **(b)** 52 **(c)** 17 **(d)** 35

15. Models of the atom have changed over time because: 1

(a) new evidence required changes **(b)** old models were hidden
(c) the atom changed **(d)** scientists were bored

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. How is a covalent bond formed? Give one example. 2

17. State two uses of isotopes. 2

18. Differentiate between a scientific law and a theory. 2

19. Define solubility. How does it vary with temperature for a solid in water? 2

20. State the law of constant proportions with an example. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. In a compound, hydrogen and oxygen are in the ratio 1 : 8 by mass. Find the mass of oxygen in 27 g of water. 3

22. Explain how Chadwick's discovery of the neutron explained the atomic mass of atoms like helium. 3

23. Explain how Meghnad Saha's work on stars shows the use of models and laws. 3

24. How would you prepare 200 mL of a 15 per cent (v/v) solution of alcohol in water? 3

OR

State three properties of covalent compounds. 3

25. State three drawbacks of Rutherford's model that were removed by Bohr's model. 3

OR

Explain the difference between accuracy and precision with an example. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Discuss the role of models in science with three examples from chemistry or physics, and explain how models change. 5

OR

Describe the properties of true solutions, suspensions and colloids using three activities. 5

27. State Dalton's atomic theory with five postulates and mention which points were modified later. 5

OR

Describe Rutherford's gold foil experiment with the observations and the inferences for each. 5

END OF QUESTION PAPER **Total: 50 Marks**

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